

Digital Material Passport

ID DMP-METAL-007 - Version 1.0.0
Issue Date 2025-05-20 - Certificate Type EN 10204 3.1

Manufacturer	Customer	Business Transaction	
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com	Bridge Constructors Europe Ltd. Engineering Plaza 456 75001 Paris FR procurement@bridgeconstructors.example.com	Order	PO-98765 · 20000 kg Pos. 5 · 2025-04-20
		Delivery	DN-12345 · 20000 kg Pos. 1 · 2025-05-19

Certificate Receiver
Civil Infrastructure Authority
Government Building 789
Administration Wing
1000 Brussels
BE
inspections@infra-authority.example.gov.be

Product

Product Name	Structural Steel S420N Plate	Name (EN)	1.8902
Batch ID	H-45678-01	Shape	Plate - Width 2500 mm - Thickness 25 mm - Length 12000 mm
Surface Condition	Shot blasted and primed	Product Norm	EN 10025-3 (2019)
Weight	20000 kg		EN 1090-2 (2018)
Production Date	2025-05-18		
Country of Origin	DE		

Chemical Analysis

Heat Number H-45678 - Melting Process EAF+LF+VD - Casting Method ContinuousCasting - Casting Date 2025-05-17 - Sample Location Ladle

Symbol	C	Mn	Si	P	S	Nb	V	Ti	CEV
Unit	%	%	%	%	%	%	%	%	%
Min		1							
Max	0.22	1.7	0.6	0.025	0.02	0.05	0.05	0.03	0.48
Actual	0.16	1.38	0.32	0.016	0.008	0.022	0.034	0.009	0.4

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.4%

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Tensile Strength ¹	Rm	548 / 550 / 552 MPa - Average 550 - Min 548 - Max 552	520 MPa	680 MPa	EN ISO 6892-1	✓
Yield Strength ¹	ReH	438 / 440 / 442 MPa - Average 440 - Min 438 - Max 442	420 MPa		EN ISO 6892-1	✓
Elongation after fracture 3 specimens tested, gauge length 5.65#S ₀ , transverse to rolling direction	A	20.5 / 21 / 21.5 % - Average 21 - Min 20.5 - Max 21.5	19 %		EN ISO 6892-1	✓
Charpy V-notch Impact Energy 3 specimens tested at -20°C, transverse to rolling direction	KV	63 / 65 / 67 J - Average 65 - Min 63 - Max 67 - Std Dev 2	40 J		EN ISO 148-1	✓

¹ 3 specimens tested at 20°C, transverse to rolling direction

Supplementary Tests

Test	Result / limits	Method	Status
Ultrasonic Testing Class S2E2	Yes - No recordable indications exceeding acceptance criteria	EN 10160	✓
Through-thickness Properties	Z25	EN 10164	✓
Weldability	Yes - Satisfactory welding properties	Internal Method based on EN ISO - 15614-1	✓

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10025-3:2019 and EN 10204:2004 type 3.1. The product complies with the Construction Products Regulation (EU) No 305/2011 and is suitable for use in structural applications according to EN 1090-2:2018, up to and including Execution Class EXC3.



